

capital mobility, which occurs when there are restrictions on cross-border capital flows or when domestic and foreign bonds are imperfect substitutes, there can be a difference between i and i_f . However, here we will only consider the case of perfect capital mobility.

5.3 MONETARY AND FISCAL POLICIES UNDER FLEXIBLE EXCHANGE RATE

When a country has a flexible exchange rate regime, its exchange rate is determined by demand and supply in the foreign exchange market. We consider the case of a *pure float*, where there is no official intervention in currency markets and hence no role of foreign exchange reserves, with BOP equilibrium achieved via exchange rate fluctuations.

5.3.1 Fiscal Policy under Flexible Exchange Rates

Consider the open economy IS-LM model with perfect capital mobility and suppose the government follows expansionary fiscal policy in this set up as in Fig. 5.2.

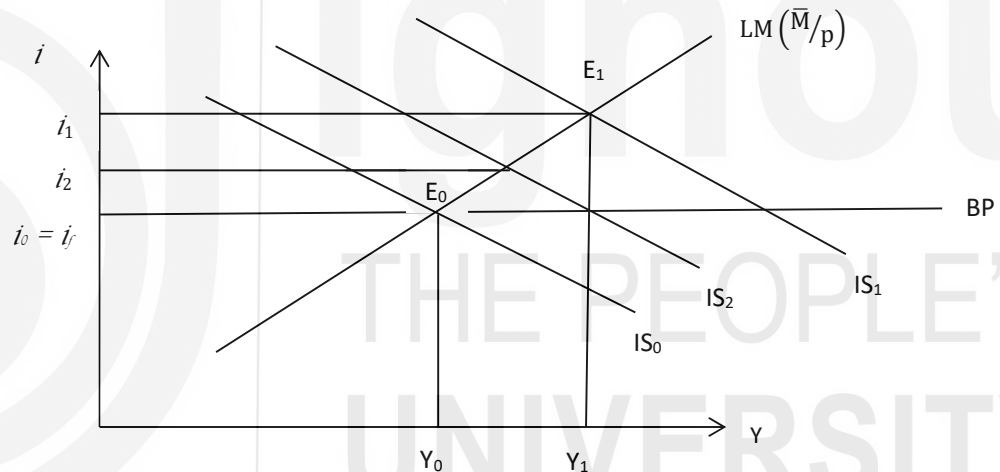


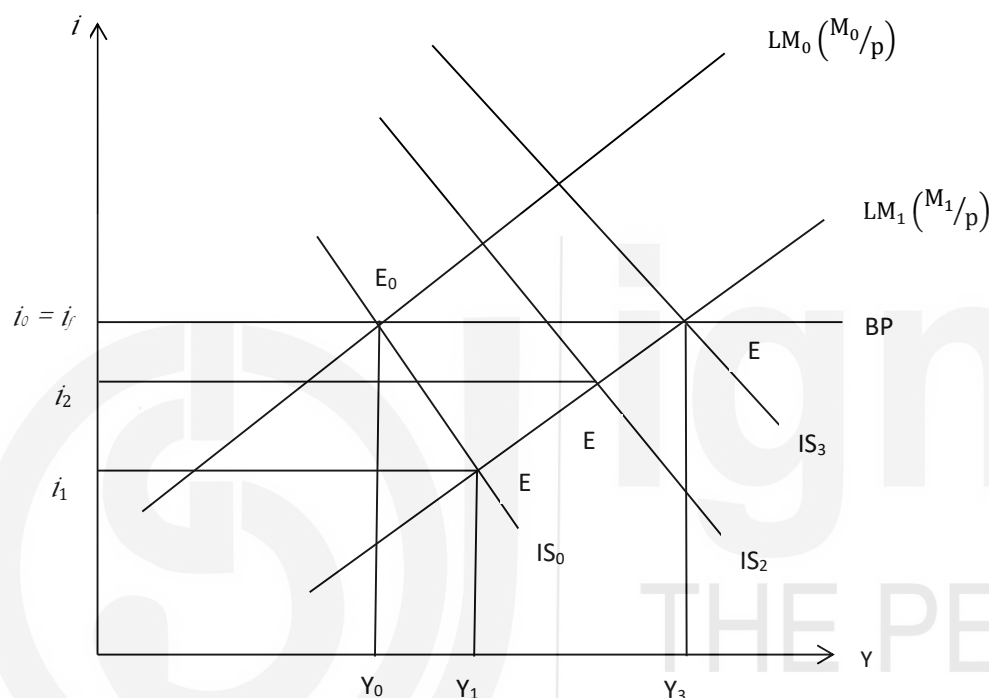
Fig. 5.2: Expansionary Fiscal Policy under Flexible Exchange Rates

Initially the economy is at E_1 (see Fig. 5.2), where goods and money markets are in equilibrium and there is BOP equilibrium as well, with $i_0 = i_f$. With fiscal expansion (or fall in tax rate t), there is an increase in G . The IS curve shifts from IS_0 to IS_1 and the interest rises from i_0 to i_1 as the economy moves to the new equilibrium E_1 . However, with i_f remaining the same, the rise in domestic interest rate makes domestic bonds relatively more attractive and there is a surge in capital inflows, that leads to a BOP surplus (excess supply of foreign exchange), causing a currency appreciation (e falls). With given prices (implying prices remaining constant), a nominal appreciation leads to a real appreciation (r falls), reducing X , raising M and worsening the trade balance (NX falls). As a result aggregate demand falls and the IS curve shifts to the left. It shifts back all the way to IS_0 , where once again $i = i_0 = i_f$ and output is back to Y_0 . Note that at any other IS curve, e.g., IS_2 , the domestic interest rate is still above i_f ($i_2 > i_f$), so e , r

and NX would fall and IS would shift further to the left. Therefore, we see that with flexible exchange rates, fiscal expansion is ineffective. It fails to raise output (the increase in output from Y_0 to Y_1 is only temporary), the trade balance worsens because of appreciation in the exchange rate (NX is lower) and the fiscal position deteriorates (G has increased or tax rate t has fallen).

5.3.2 Monetary Policy under Flexible Exchange Rates

We now examine the impact of expansionary monetary policy under a flexible exchange rate regime.



[Fig. 5.3: Expansionary Monetary Policy under Flexible Exchange Rates]

In Fig. 5.3 an increase in nominal money supply (from M_0 to M_1) shifts the LM curve outward from LM_0 to LM_1 and the economy moves from E_0 to E_1 , as domestic interest rate falls from i_0 to i_1 . At i_1 there is internal balance (output and money markets are in equilibrium) but there is external imbalance (BOP disequilibrium) that triggers capital flows. Since $i_1 < i_f$, this triggers a capital outflow, leading to a BOP deficit (excess demand for dollars) and causing a depreciation of the nominal (e rises) and real exchange rates (r rises). As a result domestic goods become relatively cheaper, X rises, M falls and NX increases, and with higher external demand, the IS curve shifts to the right. This process continues till the IS curve shifts all the way to IS_3 and the economy is at E_3 , where once again $i = i_f$ and both internal and external balance are restored. At any other point (e.g., E_2 on IS_2 , $i_2 < i_f$, so e and r would rise, NX would fall, moving the IS curve further to the right).

This shows, that in complete contrast to fiscal policy, monetary expansion is an extremely effective policy tool in an open economy with flexible exchange rates

and perfect capital mobility. It brings about currency depreciation that improves the trade balance, so that output increases from Y_0 all the way to Y_3 .

Check Your Progress 1

1. What is the effect of monetary expansion on output and interest rates in an open economy with perfect capital mobility under flexible exchange rate?

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2. Comment on the effectiveness of monetary policy in open economies in light of your answer to Question 1 above.

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3. What is the effect of fiscal contraction (cut in tax rate) on output and interest rates in an open economy with perfect capital mobility under flexible exchange rates?

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5.4 MONETARY AND FISCAL POLICIES UNDER FIXED EXCHANGE RATE

With fixed exchange rates, exchange rate fluctuations are ruled out and the central bank plays a very important role as it holds foreign exchange reserves and intervenes in foreign exchange markets to maintain the fixed parity.

5.4.1 Fiscal Policy under Fixed Exchange Rates

We first consider the impact of expansionary fiscal policy under a fixed exchange rate system, where $e = e^*$ is the official parity (Fig. 5.4).

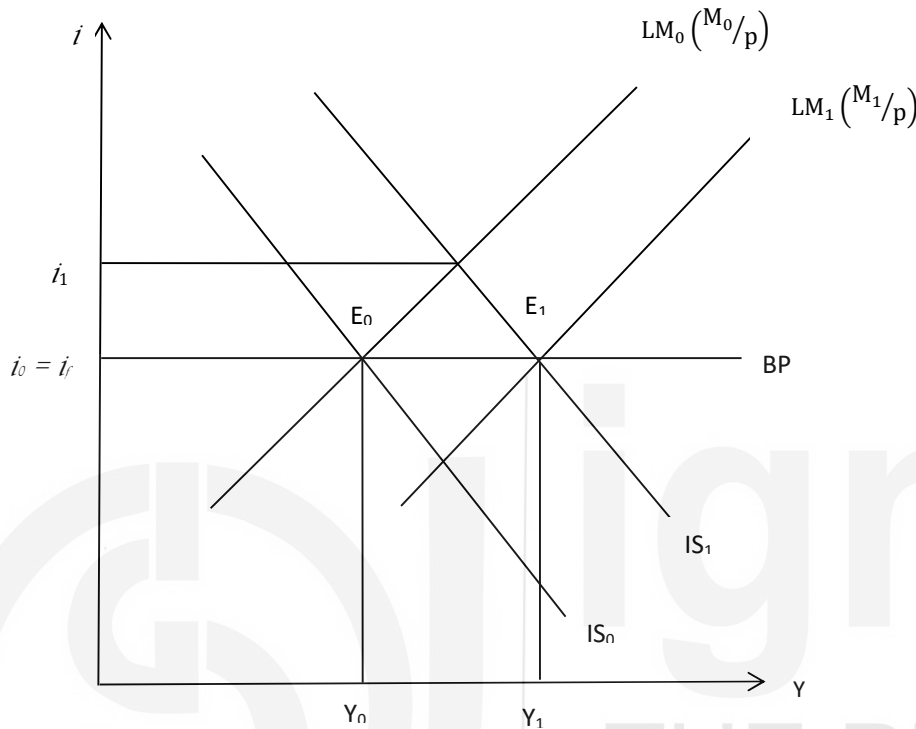


Fig. 5.4: Expansionary Fiscal Policy under Fixed Exchange Rates

Suppose there is fiscal expansion, either through an increase in government expenditure or via a cut in taxes. The IS curve shifts out from IS_0 to IS_1 and the interest rate rises to i_1 (see Fig. 5.4). With higher domestic interest rates ($i_1 > i_f$), there is a surge in capital inflows as there is BOP surplus and an excess supply of dollars (foreign exchange). This creates pressure for currency appreciation, but the central bank is committed to maintaining a fixed rate. Therefore it intervenes and buys dollars in order to mop up the excess supply. As a result foreign exchange reserves increase, leading to an increase in money supply, which shifts the LM curve outward. This process continues as long as $i > i_f$ and it stops only when LM shifts to LM_1 , where money supply has increased to M_1 ($M_1 > M_0$) and the economy reaches E_1 where $i = i_f$, output has increased to Y_1 and there is no crowding out.

Therefore, fiscal policy is extremely effective under a system of fixed exchange rates, as change in foreign exchange reserves brings about monetary expansion to accommodate the fiscal expansion. This is in complete contrast to the case of flexible exchange rates under which fiscal policy was quite ineffective.

5.4.2 Monetary Policy under Fixed Exchange Rates

Now we consider the case of a monetary expansion under a system of fixed exchange rates (Fig. 5.5).

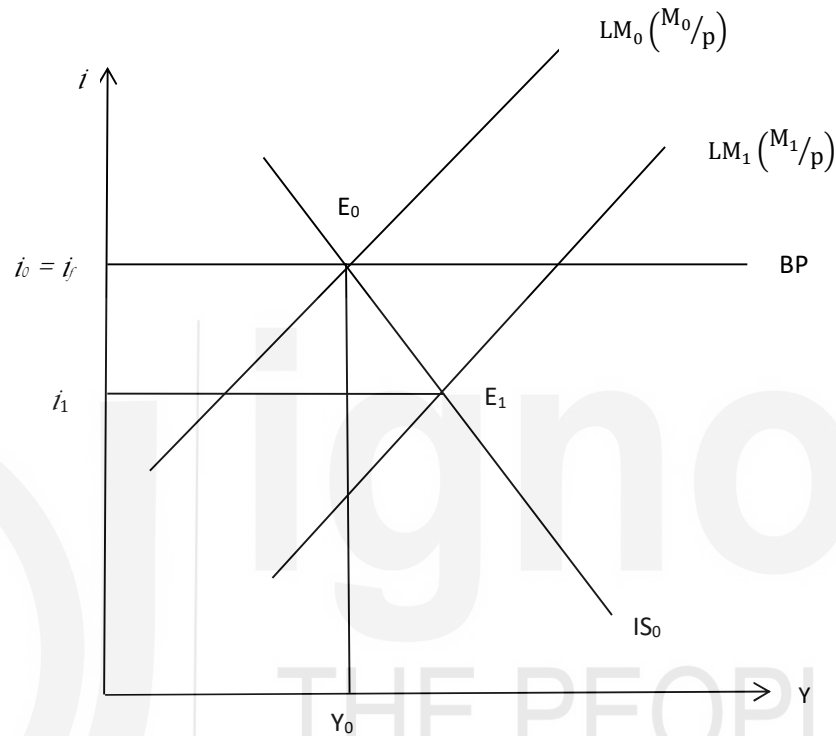


Fig. 5.5: Expansionary Monetary Policy under Fixed Exchange Rates

Suppose the central bank engages in an open market purchase of bonds in a bid to increase domestic money supply. This shifts the LM curve from LM_0 to LM_1 (since money supply increases from M_0 to M_1) and the economy moves to E_1 , where the interest rate $i_1 < i_f$ (see Fig. 5.5). Due to lower domestic rates of interest, investors would move to foreign bonds, triggering capital outflows and giving rise to a BOP deficit and excess demand for dollars. This creates pressure for currency depreciation, but the central bank is committed to maintain a fixed parity. Therefore it intervenes to sell dollars against rupees, leading to a fall in foreign exchange reserves and contraction in money supply. As money supply falls, the LM curve shifts to the left of LM_1 and this process continues as long as $i < i_f$. Ultimately the economy is back to E_0 , with money supply falling back to its initial level (M_0), so that $i = i_f$ and output is back to its initial level Y_0 .

Clearly with fixed exchange rates and perfect capital mobility, money supply becomes ineffective as a policy tool as monetary expansion has no effect on output in the short run.

In fact the above discussion indicates that money supply becomes *endogenous* with fixed exchange rates and perfect capital mobility, in the sense that it passively adjusts to changes in the foreign exchange reserves as the central bank intervenes in foreign currency markets to defend the fixed parity. That is, since the central bank wants to hold the exchange rate constant, it has to intervene to keep the exchange rate from depreciating or appreciating, so foreign exchange reserves and hence money supply adjusts accordingly.

Our discussion also highlights the fact that it is impossible to simultaneously have the following three features in a single policy regime, viz.: (a) perfect capital mobility; (b) independent monetary policy; and (c) fixed exchange rates. This is also known as the '*policy trilemma*'. In the presence of perfect capital mobility and *fixed* exchange rates, money supply becomes endogenous and the Central Bank's attempt to pursue an independent monetary policy is not effective as the Central Bank is committed to maintaining a fixed parity. Whereas, in the case of *flexible* exchange rates, perfect capital mobility and independent monetary policy do indeed go together, as we saw that monetary policy is very effective in this case. Countries may adopt fixed exchange rates and yet retain monetary policy independence only with restrictions on international capital flows, i.e., in the absence of perfect capital mobility. In this case, interest rate differentials between countries would persist owing to restrictions on capital mobility.

Check Your Progress 2

1. What is the effect of fiscal contraction (cut in tax rate) on output and interest rates in an open economy with perfect capital mobility under fixed exchange rate?

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2. Comment on the effectiveness of fiscal policy in open economies in light of your answer to question 1 above.

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3. (a) What is the effect of monetary contraction (sale of bonds by central bank) on output and interest rate in an open economy with perfect capital mobility under (i) fixed exchange rates, and (ii) flexible exchange rates?

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- (b) Explain the concept of ‘endogeneity’ of money supply.

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- (c) Comment on the effectiveness of monetary policy in open economies in light of your answer in part (a) above.

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4. Use the open economy IS-LM model to explain the concept of ‘policy trilemma’.

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5.5 LET US SUM UP

In this Unit we discussed the open economy IS-LM model wherein we saw that international trade affects aggregate demand in an economy, while capital flows play an important role in shaping policy outcomes through their impact on interest rates, exchange rates and money supply. In particular we saw that with fixed exchange rates and perfect capital mobility, money supply becomes endogenous, i.e., countries cannot conduct independent monetary policy. Fiscal

policy is an effective policy tool when the exchange rate is fixed; however, with flexible exchange rates and perfect capital mobility, fiscal expansion becomes ineffective as it has no effect on output and leads to a worsening of the trade balance. Finally, the effectiveness of monetary policy under flexible exchange rates demonstrates that countries can have at most any *two* of the three policy choices of having *independent monetary policy*, *fixed exchange rates* and *perfect capital mobility*.

5.6 ANSWER/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Refer to Sub-section 5.3.1 and Fig. 5.3.
- 2) Refer to Sub-section 5.3.1.
- 3) Refer to Sub-sections 5.2.2 and 5.2.3 and Fig. 5.1

Check Your Progress 2

- 1) Refer to Sub-section 5.4.1 and Fig. 5.4.
- 2) Refer to Sub-section 5.4.1 and Fig. 5.4.
- 3) a) Refer to Sub-section 5.3.2 and 5.4.2, and Fig. 5.3 and Fig. 5.5.
b) Refer to Sub-section 5.4.2, and Fig. 5.5.
c) Refer to Sub-sections 5.3.2 and 5.4.2, and Fig. 5.3 and Fig. 5.5.
- 4) Refer to Sub-section 5.4.2.